UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 001 · Origin of electrode potential, cell potential · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·10	Deck 001, slide 10 (rendered .pptx)
T001 00:08:22	AV-summary transcript 001, segment start
ESA Jan-25 s2 2b	Past paper: Jan 2025 set 2, question 2.b
ESA Jul-23a 4d	Past paper: Jul 2023 (first UE22 paper), question 4.d

Contents

1. Part A · Formula sheet
2. 2 · Electrode potential and its origin: the Helmholtz double layer
3. 3 · Standard electrode potential, electrochemical cell and cell notation
4. 4 · Cell potential, standard cell potential and free energy
5. Part B · Symbol glossary

A Formula sheet

2 Electrode potential and its origin: the Helmholtz double layer

E2.1 · The two competing electrode reactions

ESA ★★

$$\underbrace{\mathrm{M} \rightleftharpoons \mathrm{M}^{n+} + n\mathrm{e}^-}_{\text{ionisation (oxidation)}} \qquad \underbrace{\mathrm{M}^{n+} + n\mathrm{e}^- \rightleftharpoons \mathrm{M}}_{\text{deposition (reduction)}}$$

GIVES	The origin of the electrode potential: the faster reaction decides the sign of the charge on the metal.
APPLIES	A metal M in contact with a solution of its own ions.
UNITS	—
NOTE	Ionisation faster: metal negative (case I). Deposition faster: metal positive (case II).

S001·5 · S001·6 · S001·7

3 Standard electrode potential, electrochemical cell and cell notation

E3.1 · Standard conditions for E°

ESA ★★

$$E^\circ = E \text{ at } [\mathrm{M^{n+}}] = 1\ \mathrm{M}, \quad T = 298\ \mathrm{K}, \quad p_{gas} = 1\ \mathrm{atm}$$

GIVES	The standard electrode potential, which depends only on the nature of the electrode.
APPLIES	Comparing electrodes; tabulated in the electrochemical series.
UNITS	V
NOTE	Tabulated as standard reduction potentials.

S001·8 · T001 00:11:27

E3.2 · Cell notation (Daniell cell)

ESA ★★★

$$\underbrace{\mathrm{Zn(s)\,|\,Zn^{2+}(1\,M)}}_{\text{anode (LHS)}}\ ||\ \underbrace{\mathrm{Cu^{2+}(1\,M)\,|\,Cu(s)}}_{\text{cathode (RHS)}}$$

GIVES	A compact representation of the cell.
APPLIES	Writing any cell: anode first, salt bridge in the middle.
UNITS	—
NOTE	The slide writes / and // for | and ||; both are accepted.

S001·8 · T001 00:12:38

4 Cell potential, standard cell potential and free energy

E4.1 · Cell potential

ESA ★★★

$$E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$$

GIVES	The emf of the cell from the two electrode (reduction) potentials.
APPLIES	Every cell; with E° values it gives E°_{cell}.
UNITS	V
NOTE	Use reduction potentials for both electrodes; do not change the sign of the anode value before subtracting.

S001·9 · T001 00:15:37

E4.2 · Standard cell potential

ESA ★★★

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

GIVES The emf when all concentrations are 1 M, gases at 1 atm, $T = 298$ K.

APPLIES First step of every cell numerical (then Nernst, class 002).

UNITS V

NOTE Daniell: $0.34 - (-0.76) = 1.10$ V.

S001·9 · T001 00:15:20

E4.3 · Free energy and cell potential

ESA ★★★

$$\Delta G = -nFE_{cell} \qquad \Delta G^\circ = -nFE^\circ_{cell}$$

GIVES The free-energy change of the cell reaction (= minus the maximum electrical work).

APPLIES Deciding spontaneity; converting emf to energy.

UNITS ΔG in J mol^{-1} (E in V, F in C mol^{-1})

NOTE n = moles of electrons transferred in the balanced cell reaction; $F = 96500$ C mol^{-1}. The standard form $\Delta G^\circ = -nFE^\circ$ is stated in the class 002 derivation.

S001·9 · T001 00:16:25 · T001 00:21:24

E4.4 · Spontaneity criterion

ESA ★★★

$$E_{cell} > 0 \;\Leftrightarrow\; \Delta G < 0 \text{ (spontaneous)} \qquad E_{cell} < 0 \;\Leftrightarrow\; \Delta G > 0 \text{ (non-spontaneous)}$$

GIVES Whether the cell reaction as written proceeds on its own.

APPLIES After computing E_{cell}.

UNITS —

NOTE Galvanic cells run on reactions with $E_{cell} > 0$; electrolytic cells force reactions with $E_{cell} < 0$.

S001·9 · T001 00:16:39

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential	Potential at a metal–solution interface	V	T2
E°	Standard electrode potential	Standard reduction potential (1 M, 298 K, 1 atm)	V	E3.1, E4.2
E_{cell}	Cell potential (emf)	$E_{cathode} - E_{anode}$	V	E4.1, E4.3
E°_{cell}	Standard cell potential	$E^\circ_{cathode} - E^\circ_{anode}$	V	E4.2
E_{rhs}, E_{lhs}	Right / left electrode	Cathode / anode in the cell notation	V	E4.1
$\Delta G,\ \Delta G^\circ$	Gibbs free-energy change	Of the cell reaction (standard: ΔG°)	J mol^{-1}	E4.3
n	Electrons transferred	Moles of electrons per mole of overall reaction; in $\mathrm{M^{n+}}$ it is the ion charge (the same n for a single electrode)	—	E2.1, E4.3
F	Faraday constant	Charge on one mole of electrons, 96500 (96485)	C mol^{-1}	E4.3
M, $\mathrm{M^{n+}}$	Metal and its ion	Electrode metal and its ions in solution	—	E2.1
| , ||	Phase boundary, salt bridge	Cell notation (slides: / and //)	—	E3.2

Remark. n means the charge of the ion in $\mathrm{M^{n+}}$ and the number of electrons transferred; for a single electrode they coincide, for a cell n is the electron count of the balanced overall reaction (6 for $\mathrm{3Fe + 2Au^{3+}}$).